

Reduction of Quadratic Form to Canonical Form

By Orthogonal Reduction Method

Problem:

Reduce the quadratic form

$$Q = 6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 - 2x_2x_3 + 4x_3x_1$$

to canonical form by orthogonal reduction.

Solution:

Step 1: Form the Symmetric Matrix

Write the quadratic form in matrix notation:

$$Q = \mathbf{x}^T \mathbf{A} \mathbf{x}$$

where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

Identify coefficients:

For x_i^2 terms: coefficient goes directly on diagonal

For $x_i x_j$ terms (where $i \neq j$): coefficient is split equally as $\frac{a_{ij}}{2}$ in positions (i,j) and (j,i)

Given quadratic form:

$$Q = 6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 - 2x_2x_3 + 4x_3x_1$$

Coefficient extraction:

- $a_{11} = 6$ (coefficient of x_1^2)
- $a_{22} = 3$ (coefficient of x_2^2)
- $a_{33} = 3$ (coefficient of x_3^2)
- $a_{12} = a_{21} = \frac{-4}{2} = -2$ (coefficient of x_1x_2 divided by 2)
- $a_{23} = a_{32} = \frac{-2}{2} = -1$ (coefficient of x_2x_3 divided by 2)
- $a_{13} = a_{31} = \frac{4}{2} = 2$ (coefficient of x_1x_3 divided by 2)

Symmetric Matrix:

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

Step 2: Find the Characteristic Equation

The characteristic equation is:

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

Expand the determinant using first row:

$$(6-\lambda) \begin{vmatrix} 3-\lambda & -1 \\ -1 & 3-\lambda \end{vmatrix} - (-2) \begin{vmatrix} -2 & -1 \\ 2 & 3-\lambda \end{vmatrix} + 2 \begin{vmatrix} -2 & 3-\lambda \\ 2 & -1 \end{vmatrix} = 0$$

Calculate each 2×2 determinant:

First determinant:

$$\begin{vmatrix} 3-\lambda & -1 \\ -1 & 3-\lambda \end{vmatrix} = (3-\lambda)^2 - 1 = 9 - 6\lambda + \lambda^2 - 1 = \lambda^2 - 6\lambda + 8$$

Second determinant:

$$\begin{vmatrix} -2 & -1 \\ 2 & 3-\lambda \end{vmatrix} = -2(3-\lambda) - (-1)(2) = -6 + 2\lambda + 2 = 2\lambda - 4$$

Third determinant:

$$\begin{vmatrix} -2 & 3-\lambda \\ 2 & -1 \end{vmatrix} = (-2)(-1) - (3-\lambda)(2) = 2 - 6 + 2\lambda = 2\lambda - 4$$

Combine:

$$(6-\lambda)(\lambda^2 - 6\lambda + 8) + 2(2\lambda - 4) + 2(2\lambda - 4) = 0$$

$$(6-\lambda)(\lambda^2 - 6\lambda + 8) + 4(2\lambda - 4) = 0$$

$$(6-\lambda)(\lambda^2 - 6\lambda + 8) + 8\lambda - 16 = 0$$

Expand $(6-\lambda)(\lambda^2 - 6\lambda + 8)$:

$$6\lambda^2 - 36\lambda + 48 - \lambda^3 + 6\lambda^2 - 8\lambda = 0$$

$$-\lambda^3 + 12\lambda^2 - 44\lambda + 48 + 8\lambda - 16 = 0$$

$$-\lambda^3 + 12\lambda^2 - 36\lambda + 32 = 0$$

$$\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$$

Solve by trial method:

Try $\lambda = 2$: $8 - 48 + 72 - 32 = 0 \checkmark$

So $(\lambda - 2)$ is a factor.

Factor out $(\lambda - 2)$:

$$\lambda^3 - 12\lambda^2 + 36\lambda - 32 = (\lambda - 2)(\lambda^2 - 10\lambda + 16)$$

Factor the quadratic:

$$\lambda^2 - 10\lambda + 16 = (\lambda - 2)(\lambda - 8)$$

Eigenvalues:

$$\boxed{\lambda_1 = 2, \quad \lambda_2 = 2, \quad \lambda_3 = 8}$$

Step 3: Find Eigenvectors

For $\lambda_1 = 2$ (repeated eigenvalue):

Solve $(A - 2I)v = 0$:

$$A - 2I = \begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix}$$

Set up the system:

$$\begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Row reduce:

$$R_2 \rightarrow R_2 + \frac{1}{2}R_1:$$

$$\begin{bmatrix} 4 & -2 & 2 \\ 0 & 0 & 0 \\ 2 & -1 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - \frac{1}{2}R_1:$$

$$\begin{bmatrix} 4 & -2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

This gives: $4v_1 - 2v_2 + 2v_3 = 0$ or $2v_1 - v_2 + v_3 = 0$

Choose independent vectors:

Let $v_2 = 2, v_3 = 0$: Then $2v_1 - 2 = 0 \Rightarrow v_1 = 1$

First eigenvector: $v_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$

Let $v_2 = 0, v_3 = 2$: Then $2v_1 + 2 = 0 \Rightarrow v_1 = -1$

Second eigenvector: $v_2 = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$

Normalize the eigenvectors:

$$u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{1^2+2^2+0^2}} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$u_2 = \frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{1+0+4}} \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

For $\lambda_3 = 8$:

Solve $(A - 8I)v = 0$:

$$A - 8I = \begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix}$$

Row reduce:

$R_1 \rightarrow -\frac{1}{2}R_1$:

$$\begin{bmatrix} 1 & 1 & -1 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix}$$

$R_2 \rightarrow R_2 + 2R_1$ and $R_3 \rightarrow R_3 - 2R_1$:

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & -3 & -3 \\ 0 & -3 & -3 \end{bmatrix}$$

From second row: $-3v_2 - 3v_3 = 0 \Rightarrow v_2 = -v_3$

From first row: $v_1 + v_2 - v_3 = 0 \Rightarrow v_1 = v_3 - v_2 = 2v_3$

Choose $v_3 = 1$:

Then $v_2 = -1$ and $v_1 = 2$

Third eigenvector: $v_3 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$

Normalize:

$$u_3 = \frac{1}{\sqrt{4+1+1}} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{6}} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

Normalized Eigenvectors:

$$u_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad u_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \quad u_3 = \frac{1}{\sqrt{6}} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

Step 4: Form the Orthogonal Matrix P

The orthogonal matrix **P** consists of normalized eigenvectors as columns:

$$P = [u_1 \quad u_2 \quad u_3]$$

$$P = \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & 0 & -\frac{1}{\sqrt{6}} \\ 0 & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{6}} \end{bmatrix}$$

Verification that P is orthogonal:

For an orthogonal matrix: $P^T P = I$

Check that columns are orthonormal:

- $u_1 \cdot u_1 = \frac{1}{5}(1 + 4 + 0) = 1 \checkmark$
- $u_2 \cdot u_2 = \frac{1}{5}(1 + 0 + 4) = 1 \checkmark$
- $u_3 \cdot u_3 = \frac{1}{6}(4 + 1 + 1) = 1 \checkmark$
- $u_1 \cdot u_2 = \frac{1}{5}(-1 + 0 + 0) = -\frac{1}{5} + \frac{1}{5} = 0 \checkmark$

Step 5: Apply the Transformation

The transformation is:

$$x = Py$$

where $y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ are the new variables.

The quadratic form becomes:

$$Q = \mathbf{x}^T \mathbf{A} \mathbf{x} = (\mathbf{P} \mathbf{y})^T \mathbf{A} (\mathbf{P} \mathbf{y}) = \mathbf{y}^T (\mathbf{P}^T \mathbf{A} \mathbf{P}) \mathbf{y}$$

Since $\mathbf{P}$ is the matrix of eigenvectors:

$$\mathbf{P}^T \mathbf{A} \mathbf{P} = \mathbf{D} = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

Canonical Form:

$$Q = 2y_1^2 + 2y_2^2 + 8y_3^2$$

Step 6: Transformation Equations

The relationship between old and new variables:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & 0 & -\frac{1}{\sqrt{6}} \\ 0 & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

Explicitly:

$$x_1 = \frac{y_1}{\sqrt{5}} - \frac{y_2}{\sqrt{5}} + \frac{2y_3}{\sqrt{6}}$$

$$x_2 = \frac{2y_1}{\sqrt{5}} - \frac{y_3}{\sqrt{6}}$$

$$x_3 = \frac{2y_2}{\sqrt{5}} + \frac{y_3}{\sqrt{6}}$$

Step 7: Nature of the Quadratic Form

Analysis of eigenvalues:

- All eigenvalues are positive: $\lambda_1 = 2 > 0$, $\lambda_2 = 2 > 0$, $\lambda_3 = 8 > 0$
- Therefore, the quadratic form is **POSITIVE DEFINITE**

Properties:

- **Index:** $p = 3$ (number of positive eigenvalues)
- **Signature:** $p - q = 3 - 0 = 3$
- **Rank:** 3 (all eigenvalues non-zero)

Final Summary

Complete Solution:

1. Original Quadratic Form:

$$Q = 6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 - 2x_2x_3 + 4x_3x_1$$

2. Matrix Representation:

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

3. Eigenvalues:

$$\lambda_1 = 2, \quad \lambda_2 = 2, \quad \lambda_3 = 8$$

4. Canonical Form:

$$Q = 2y_1^2 + 2y_2^2 + 8y_3^2$$

5. Orthogonal Matrix:

$$P = \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{6}} \\ \frac{2}{\sqrt{5}} & 0 & -\frac{1}{\sqrt{6}} \\ 0 & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{6}} \end{bmatrix}$$

6. Transformation:

$$\mathbf{x} = P\mathbf{y} \quad \text{or} \quad \mathbf{y} = P^T\mathbf{x}$$

7. Nature: The quadratic form is **POSITIVE DEFINITE**

Key Points to Remember:

- In the symmetric matrix, off-diagonal elements are half the coefficients of cross-product terms
- Eigenvalues of the matrix become coefficients in the canonical form
- The orthogonal matrix P consists of normalized eigenvectors as columns
- The transformation $x = Py$ eliminates cross-product terms
- All positive eigenvalues indicate a positive definite form